

DEPLOYMENT-ALIGNED EVALUATION WITH TEMPORAL AND GEOGRAPHIC HOLDOUTS

A Cross-Case Empirical Study of Forecasting and Semantic Segmentation

Herislei Pimentel

Usinas Siderurgicas de Minas Gerais S.A. - Usiminas, Brazil (research conducted independently)

ORCID: 0009-0001-0692-4015 | herislei@gmail.com

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Abstract

Machine-learning systems are often evaluated with data partitions that do not match the structure of intended deployment. This study examines how deployment-aligned holdouts constrain the generalization claims that can be supported across two heterogeneous applied-learning tasks. The paper does not estimate a formal magnitude of distribution shift and does not compare structured holdouts against random splitting; instead, it asks what evidence is defensible when time or geography is used as the unit of separation and the final holdout is excluded from development. Case I uses 17,379 hourly Capital Bikeshare observations (2011-2012) under a conservative monthly temporal protocol. Among the reported forecasting configurations, a weather-enhanced Histogram Gradient Boosting model produced the lowest observed errors (MAE 36.59, RMSE 61.88, RMSLE 0.461) on 5,919 evaluation observations, a 52.6% lower MAE than the historical baseline under the same protocol. The preserved notebook also shows lower HGB-weather MAE than the baseline in every plotted monthly evaluation window. Case II uses an author-defined cross-city split of the Inria Aerial Image Labeling dataset: Austin, Chicago, and Kitsap for training, Tyrol-W for validation, and labeled Vienna imagery as a held-out final evaluation city. A deterministic seed-42 U-Net/ResNet-34 execution achieved mean IoU 0.588, Dice 0.716, Precision 0.635, Recall 0.886, and Boundary F1 0.297 on 576 sampled Vienna patches. A sensitivity analysis restricted to the 543 patches containing building pixels produced similar region metrics, and retained qualitative failure cases expose substantial false positives and boundary errors. The contribution is a bounded claim-evidence reporting discipline supported by preserved computational artifacts, not a new learning algorithm or a universal robustness result.

Keywords: deployment-aligned evaluation; structured holdout; temporal validation; geographic generalization; time-series forecasting; semantic segmentation; data leakage; failure analysis; reliable machine learning

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1. Introduction

Applied machine-learning performance is meaningful only relative to the data-generating and deployment conditions represented by evaluation. Training and future-use data can differ across time, location, acquisition system, population, or operating regime [1-4]. These mismatches motivate work on dataset shift and out-of-distribution generalization, but the present study does not estimate a statistical divergence or claim that one measured shift magnitude caused the observed results. Its narrower concern is evaluation design: whether the holdout structure corresponds to the deployment question being claimed.

When observations are dependent in time or space, naive random partitions can answer a different question from future-time or new-location deployment. Temporal forecasting studies therefore distinguish fixed, rolling, expanding, and repeated holdout designs [5,17,18], while spatial prediction literature warns that ordinary cross-validation can be optimistic when spatial dependence spans training and test samples [6,19]. The Inria benchmark was explicitly motivated by cross-city generalization [7]. These literatures motivate structured holdouts, but they do not imply that the specific protocols used here are universally optimal.

The present work examines two applied machine-learning cases with deliberately different modalities. Case I evaluates bicycle-demand forecasting under a conservative temporal holdout protocol. Case II evaluates building-footprint segmentation under an author-defined city-held-out split using labeled Inria imagery. The tasks are paired not because they share targets or algorithms, but because each permits the same question to be asked at a different structural level: what generalization statement is supported when the final holdout is defined by the deployment-relevant unit (time or city) and excluded from development?

The paper's contribution is methodological synthesis and empirical documentation rather than algorithmic novelty. It (1) makes the mapping from deployment question to holdout unit explicit; (2) reports two applications with

leakage controls and multiple diagnostics; (3) uses a claim-evidence matrix to state the maximum inference supported by each experiment; and (4) documents limitations that would be hidden by a single headline metric. This reporting discipline is not claimed as a new robustness algorithm, a theorem, or a substitute for dedicated counterfactual experiments.

Because no random-split counterfactual, rotating city holdout, or multi-seed uncertainty study is available in the preserved outputs, the manuscript does not quantify optimism from conventional splitting and does not claim universal temporal or geographic robustness. Version 2.3 strengthens reproducibility by linking the manuscript to a public GitHub repository containing the deterministic notebook, plain-Python export, documented data-splitting logic, model settings, random-state controls, and evaluation definitions used for the reported results. The remaining counterfactual experiments are identified as future work rather than inferred from unavailable data.

2. Related Work

2.1 Dataset shift and out-of-distribution generalization

Dataset shift has long been recognized as a mismatch between the distributions that generate training and test data [1]. More recent benchmarks such as WILDS formalize naturally occurring shifts across time, geography, institutions, cameras, and other environments, showing that in-distribution performance can substantially overstate performance on shifted target data [2]. Studies of natural image distribution shift similarly show that robustness to synthetic perturbations does not necessarily transfer to naturally occurring changes in the data-generating process [4].

Reliability under shift also depends on whether model confidence remains meaningful. Ovadia et al. found that predictive uncertainty and calibration can deteriorate as dataset shift increases, reinforcing the need to interpret confidence cautiously outside the training distribution [3]. Calibration research likewise shows that modern neural networks can be miscalibrated even when their classification accuracy is strong [14]. These findings motivate evaluation that goes beyond a single score and explicitly examines when predictions should be trusted, reviewed, or rejected.

2.2 Temporal validation and leakage control

Time-series evaluation differs from ordinary independent and identically distributed prediction because serial dependence, changing regimes, and non-stationarity complicate performance estimation. Tashman distinguishes fixed versus rolling origins, updating versus recalibration, and single versus multiple test periods [17]. Bergmeir, Hyndman, and Koo discuss conditions under which cross-validation may be valid for autoregressive prediction [5], while Cerqueira, Torgo, and Mozetic show empirically that performance-estimation method matters and that out-of-sample procedures are particularly relevant under non-stationarity [18]. These results support explicit reporting of the temporal protocol rather than treating a train/test split as a neutral implementation detail.

The forecasting case adopts a deliberately conservative variant: for each month, the late-month evaluation window is held out and its labels are not recycled into later model fits. This is not presented as a standard production rolling-origin simulation. In many deployments, previously observed labels would become available and could be incorporated into later fits. The protocol is therefore interpreted as a conservative experimental separation that preserves a strict boundary between development labels and reported evaluation outcomes. A direct comparison with expanding-window and repeated holdout designs remains necessary to determine how sensitive the numerical results are to this choice.

2.3 Geographic generalization in spatial prediction

Spatially structured data present an analogous problem. Roberts et al. show that cross-validation that ignores temporal or spatial dependence can underestimate predictive error when dependence structures are shared between training and test samples [6]. Pohjankukka et al. likewise report optimistic performance estimates from ordinary cross-validation in spatial modeling and motivate spatially structured alternatives [19]. In aerial-image analysis,

Maggiori et al. introduced the Inria benchmark specifically to study whether semantic labeling methods generalize across geographically distinct cities [7].

This study follows that generalization objective by partitioning labeled cities before patch creation. Vienna is used here as an author-defined held-out city drawn from the labeled portion of the Inria dataset; it is not presented as the benchmark's official blind test set. The resulting evidence is therefore specific to the chosen training cities, validation city, patch-sampling scheme, and held-out Vienna evaluation. It does not establish universal geographic robustness.

2.4 Models, metrics, and failure-aware evaluation

The forecasting case uses tree-based ensemble methods, including gradient boosting [11] and random forests [12], because they can represent nonlinear interactions among calendar and weather variables without requiring a single parametric functional form. A Negative Binomial generalized linear model is used separately for interpretable association diagnostics because the count outcome exhibited substantial overdispersion relative to a Poisson model.

The segmentation case uses a U-Net architecture [9] with a ResNet-34 encoder derived from residual learning [10] and AdamW optimization [13]. Pixel-overlap measures such as IoU and Dice quantify region agreement, while Precision and Recall expose false-positive versus false-negative tendencies. Boundary F1 is included because footprint delineation can remain poor even when pixel overlap is acceptable. This multi-metric view is consistent with reliability research that treats confidence, error detection, and out-of-distribution behavior as separate questions [3,14,15]. Domain-generalization research also emphasizes that model-selection strategy is part of a complete evaluation design [20], a point that is material to the limitations of the preserved experiments reported here.

3. Research Design and Unified Evaluation Logic

3.1 Research questions

The study is organized around two research questions:

- RQ1: How does a deployment-aligned holdout structure constrain the scope of the generalization claim that can be supported in temporal forecasting and geographic segmentation?
- RQ2: Which limitations or failure modes become visible when headline predictive performance is interpreted together with baselines, statistical diagnostics, class-specific metrics, boundary quality, and explicit deployment assumptions?

Research question	Primary evidence	Permitted inference
RQ1	Pre-specified temporal holdout and author-defined city-held-out evaluation	How the holdout unit bounds interpretation in each case; not the magnitude of optimism relative to random splitting.
RQ2	Baseline comparison, descriptive regression diagnostics, Precision/Recall, Boundary F1, and qualitative error categories	Which weaknesses remain visible under the stated protocols; not universal operational safety, calibration, or robustness.

Table 1. Research questions, evidence, and deliberately bounded inference.

3.2 Common evaluation principle

The two experiments use different datasets, predictive targets, model families, and metrics. Their common element is therefore not a shared algorithm but a shared evaluation logic. The deployment-relevant structure is identified first (time for forecasting; city/geography for segmentation), and the final holdout is excluded from model fitting and model selection. Performance is interpreted with multiple diagnostics, and the resulting claim is restricted to the data and protocol actually evaluated. The manuscript uses distribution-shift literature as motivation but does not estimate formal distributional divergence between development and holdout data.

This logic is illustrated in Figure 1. It is a claim-evidence sequence rather than a new robustness algorithm. The procedure does not guarantee representativeness of all future times or locations, does not detect every possible shift, and does not establish that the chosen split is optimal. Its purpose is to make the evidentiary boundary of each experiment explicit.

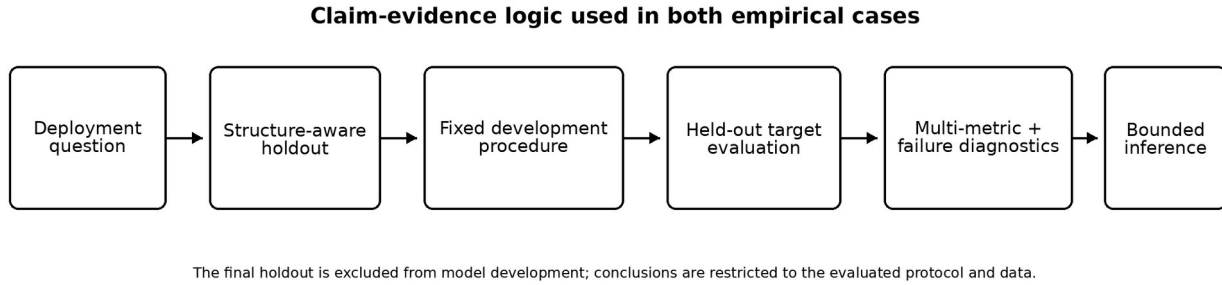


Figure 1. Claim-evidence logic used across both empirical cases. The final holdout is excluded from development, and conclusions are restricted to the evaluated protocol and data.

3.3 Case I: data and temporal partitioning

The forecasting analysis uses the hourly Bike Sharing dataset from the UCI Machine Learning Repository, associated with the Capital Bikeshare system [8,16]. The dataset contains 17,379 hourly observations from 1 January 2011 through 31 December 2012 and has no missing values in the repository version used here. Variables include calendar descriptors, weather state, temperature, humidity, wind, and bicycle rental counts.

The target is total hourly rental count. The component variables for casual and registered users are not used as predictors because their sum equals the target by construction; retaining them would constitute direct target leakage. Normalized weather variables were converted according to the dataset documentation used in the original assessment.

For each month, days 21 through month-end form the evaluation window. Training uses only observations from days 1-20 that occur before that evaluation window, and evaluation labels are not recycled into later fits. Across the complete protocol, 5,919 observations are evaluated. This design deliberately prioritizes separation over data reuse and should be interpreted as a conservative experimental protocol, not as a claim that real deployment would permanently discard labels after they become available. A standard expanding-window comparison is not available in the preserved outputs and is therefore identified as future validation work.

3.4 Case I: statistical diagnostics, models, and metrics

Temperature and apparent temperature were highly correlated ($r = 0.988$). To reduce redundancy in the descriptive inferential model, temperature was retained and apparent temperature was excluded. A Poisson generalized linear model produced Pearson dispersion of 14.37, indicating substantial overdispersion. The association analysis therefore used a Negative Binomial generalized linear model with robust standard errors. Hour and month were represented categorically, and an hour-by-working-day interaction allowed commuting patterns to vary over the day. These coefficients are interpreted as descriptive associations, not causal effects. The reported robust standard errors are not treated as a complete correction for all possible serial dependence, which further limits inferential interpretation.

Forecasting performance is reported for three fixed machine-learning configurations and a historical baseline. The preserved source notebook specifies HGB-calendar as HistGradientBoostingRegressor with Poisson loss, learning rate 0.06, 220 boosting iterations, 31 maximum leaf nodes, minimum leaf size 20, L2 regularization 1.0, and random_state=42. HGB-weather uses the same settings with 260 iterations and adds temperature, humidity, wind speed, and weather-category indicators. RF-weather-log uses 220 trees, minimum leaf size 2, max_features=0.85, random_state=42, and is fitted to $\log_{10}(\text{count})$ before inverse transformation. These configurations are explicitly recoverable from the notebook; however, the pre-notebook provenance of how those hyperparameters were

originally chosen is not documented, so the manuscript does not characterize the comparison as a pre-registered hyperparameter search.

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

$$\text{RMSE} = \sqrt{(1/n) \sum (y_i - \hat{y}_i)^2}$$

$$\text{RMSLE} = \sqrt{(1/n) \sum (\log(1+y_i) - \log(1+\hat{y}_i))^2}$$

3.5 Case II: geographic partitioning and segmentation model

The second case uses labeled imagery from the Inria Aerial Image Labeling benchmark [7]. For this study, the labeled cities were partitioned before patch creation: Austin, Chicago, and Kitsap for training; Tyrol-W for validation; and Vienna as an author-defined held-out labeled evaluation city. Vienna is part of the labeled data used to construct this experiment and is not the benchmark's official blind test set. Each 5000 x 5000 tile contributes a deterministic 4 x 4 grid of non-overlapping 512 x 512 patches. The preserved code uses start coordinates 0, 1496, 2992, and 4488 pixels on both axes, producing 16 spatially distributed patches per tile. This samples 4,194,304 pixels, approximately 16.8% of each tile area; it is therefore distributed sampling rather than full-tile coverage. Because city assignment occurs before patch creation, no city appears in more than one split.

The segmentation model is a U-Net [9] with an ImageNet-pretrained ResNet-34 encoder [10]. The estimated foreground imbalance produced a positive-class weight of 5.96. The training objective combined 0.45 weighted binary cross-entropy, 0.45 soft Dice loss, and 0.10 boundary loss. Optimization used AdamW [13] with learning rate 3e-4, weight decay 1e-4, batch size 8, a maximum of eight epochs, and model selection by validation IoU on Tyrol-W. The deterministic notebook fixes seed 42 for Python, NumPy, PyTorch, and CUDA; disables CuDNN benchmarking; enables deterministic CuDNN behavior; requests deterministic algorithms where supported; fixes the training DataLoader generator; uses num_workers=0; and resets the PyTorch RNG immediately before model initialization. Evaluation uses a fixed probability threshold of 0.50. Boundary F1 is computed with a two-pixel tolerance implemented through a 5 x 5 dilation. These controls make the reported single run reproducible in random-state design, although exact package versions from the original 2026 Colab runtime were not archived.

$$\text{IoU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$$

$$\text{Dice} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}); \quad \text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

3.6 Cross-case analytic strategy

The cases are compared analytically rather than through a pooled statistical test because their outcomes, units, error structures, and metrics are not commensurate. The comparison focuses on five dimensions: the deployment-relevant structure represented by the holdout; the unit at which separation is imposed; the leakage guard; the diagnostics used beyond aggregate performance; and the maximum defensible claim supported by the experiment.

No inference is made that the selected algorithms are universally superior, that the structured holdouts are quantitatively better than random splits in these datasets, or that a measured distribution-shift magnitude explains the results. Those claims would require dedicated counterfactual experiments and additional diagnostics. The analysis instead asks how the stated holdout design bounds the interpretation of the preserved empirical results.

4. Results

4.1 Forecasting: statistical diagnostics

Table 2 summarizes the Negative Binomial association estimates retained from the original analysis. The incident-rate ratios (IRRs) show strong associations with temperature, weather category, and year after accounting for the modeled calendar structure. Because the study is observational, these estimates are descriptive associations rather than causal effects.

Term	IRR	95% CI low	95% CI high
Temperature (+5 C)	1.184	1.174	1.195
Humidity (+10 units)	0.983	0.978	0.987
Wind (+10 units)	0.958	0.951	0.965
Weather category 2	0.940	0.926	0.953
Weather category 3	0.602	0.583	0.622
Weather category 4	0.578	0.471	0.710
Year 2012	1.605	1.587	1.624

Table 2. Negative Binomial generalized linear model association estimates.

4.2 Forecasting: walk-forward predictive performance

Among the four reported forecasting configurations, HGB-weather produced the lowest observed pooled errors under the stated conservative temporal protocol (Table 3). Its MAE of 36.592 is 40.677 rentals per hour lower than the historical baseline MAE of 77.269, equivalent to a 52.6% lower MAE under the same evaluation protocol. RF-weather-log ranked next, while HGB-calendar remained below the historical baseline. The preserved notebook additionally generated month-level MAE trajectories (Figure 2). HGB-weather is below the historical baseline in every plotted monthly evaluation window, while absolute errors vary materially over time, showing why the pooled summary should not be interpreted as a single homogeneous test condition.

Model	MAE	RMSE	RMSLE	Evaluation n
HGB-weather	36.592	61.882	0.461	5,919
RF-weather-log	39.140	65.496	0.465	5,919
HGB-calendar	47.529	82.660	0.543	5,919
Historical baseline	77.269	115.381	0.627	5,919

Table 3. Reported forecasting performance under the conservative temporal holdout protocol.

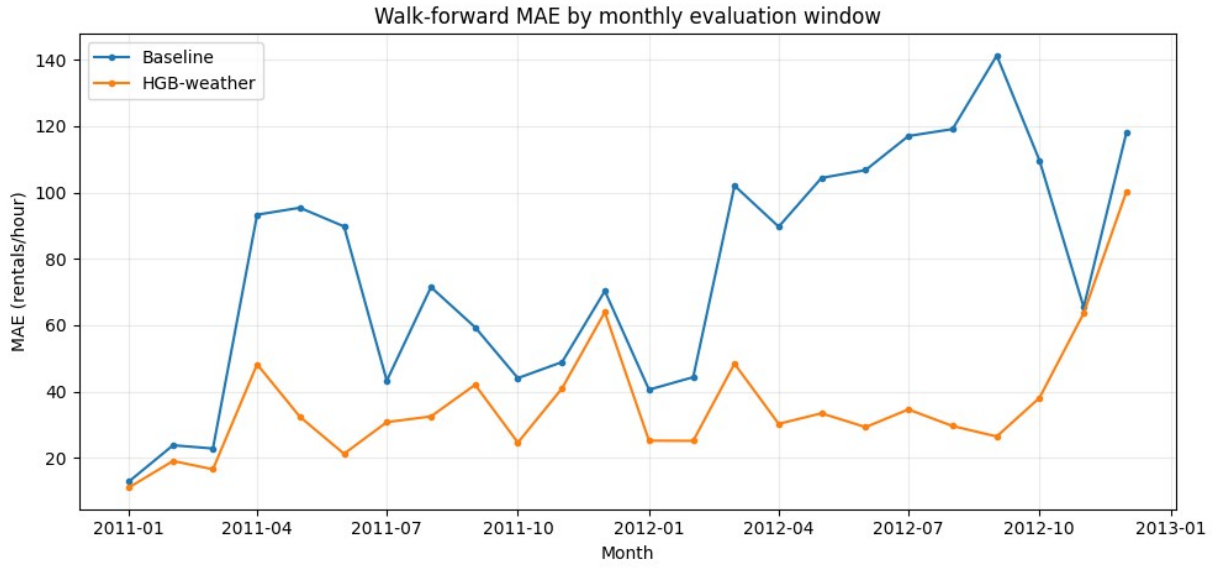


Figure 2. MAE by monthly evaluation window from the preserved forecasting notebook. The plot is an original computational output, not a reconstructed illustration.

The strongest forecasting result is conditional rather than unconditional. HGB-weather assumes that comparable weather covariates are available *ex ante*, for example from a forecast service. The experiment uses the dataset weather fields as predictors and therefore does not quantify the additional error that would arise from imperfect operational weather forecasts. HGB-calendar is consequently an important assumption-light reference point.

4.3 Segmentation: split integrity and final-test performance

Table 4 records the author-defined city-level partition used before patch generation. The design provides one held-out geographic evaluation within the labeled Inria data: three training cities, one validation city, and Vienna as the final held-out evaluation city. In the deterministic execution preserved for this manuscript, the highest Tyrol-W validation IoU was 0.5663 at epoch 6; that checkpoint was retained for the final Vienna evaluation. Figure 3 reproduces the original validation trajectory from the notebook.

Split	Cities	Tiles	Patches
Training	Austin, Chicago, Kitsap	108	1,728
Validation	Tyrol-W	36	576
Held-out evaluation	Vienna	36	576

Table 4. Author-defined city split used in the building-footprint segmentation case.

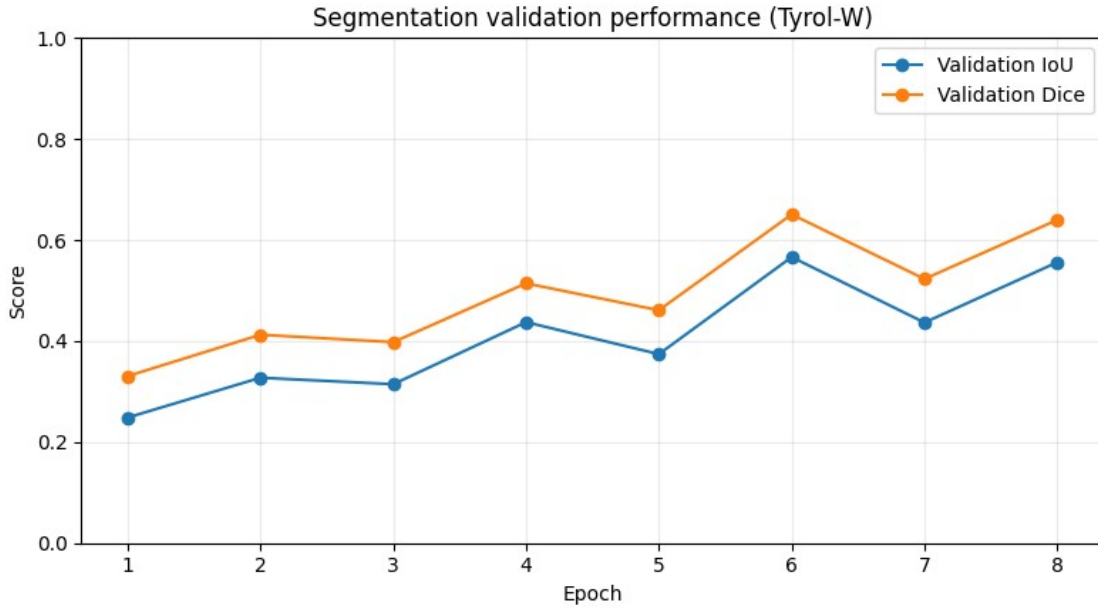


Figure 3. Tyrol-W validation IoU and Dice across the eight deterministic training epochs. The highest validation IoU in the preserved run occurred at epoch 6.

On the 576 author-defined held-out Vienna patches, the deterministic run achieved mean IoU 0.5876 and mean Dice 0.7156 (Table 5). Recall was substantially higher than Precision, and Boundary F1 was much lower than the region-overlap metrics. The reported standard deviations are patch-level dispersion, not confidence intervals for city-level generalization; spatial dependence among patches may reduce the effective amount of independent geographic information. Of the 576 patches, 543 (94.3%) contained at least one ground-truth building pixel. Restricting the diagnostic calculation to those non-empty patches produced mean IoU 0.5920 (SD 0.1581), Dice 0.7278 (SD 0.1630), Precision 0.6418 (SD 0.1594), and Recall 0.8790 (SD 0.1610). The similarity of these values to the all-patch results indicates that the reported region-overlap performance is not driven by a large set of trivial empty-background patches.

Metric	Mean	Standard deviation
IoU	0.5876	0.1955
Dice	0.7156	0.2045
Precision	0.6345	0.1979
Recall	0.8859	0.1588
Boundary F1	0.2967	0.1889

Table 5. Building-footprint segmentation metrics on the sampled patches from held-out Vienna.

4.4 Segmentation failure analysis

The metric pattern is informative because a single overlap score would conceal the nature of the errors. Recall of 0.8859 indicates that most positive building pixels in the sampled patches were identified, while Precision of 0.6345 shows a substantial false-positive burden. Boundary F1 of 0.2967 indicates weaker edge localization relative to coarse region detection. The preserved notebook also selected three non-trivial qualitative cases only among patches with at least 1% ground-truth building coverage: the lowest-IoU case (IoU 0.025), a median case (IoU 0.636), and the highest-IoU case (IoU 0.893). This filter is used only for visualization; all 576 Vienna patches remain in the quantitative metrics.

Figure 4 reproduces those original aerial image / ground-truth / prediction triplets. The lowest-IoU example demonstrates severe false-positive structure relative to a small true-building footprint; the median example shows substantial overlap but imperfect geometry; and the highest-IoU example shows close agreement on a large industrial roof. Retaining the failure case, rather than showing only favorable examples, makes the precision and boundary limitations visually auditable.

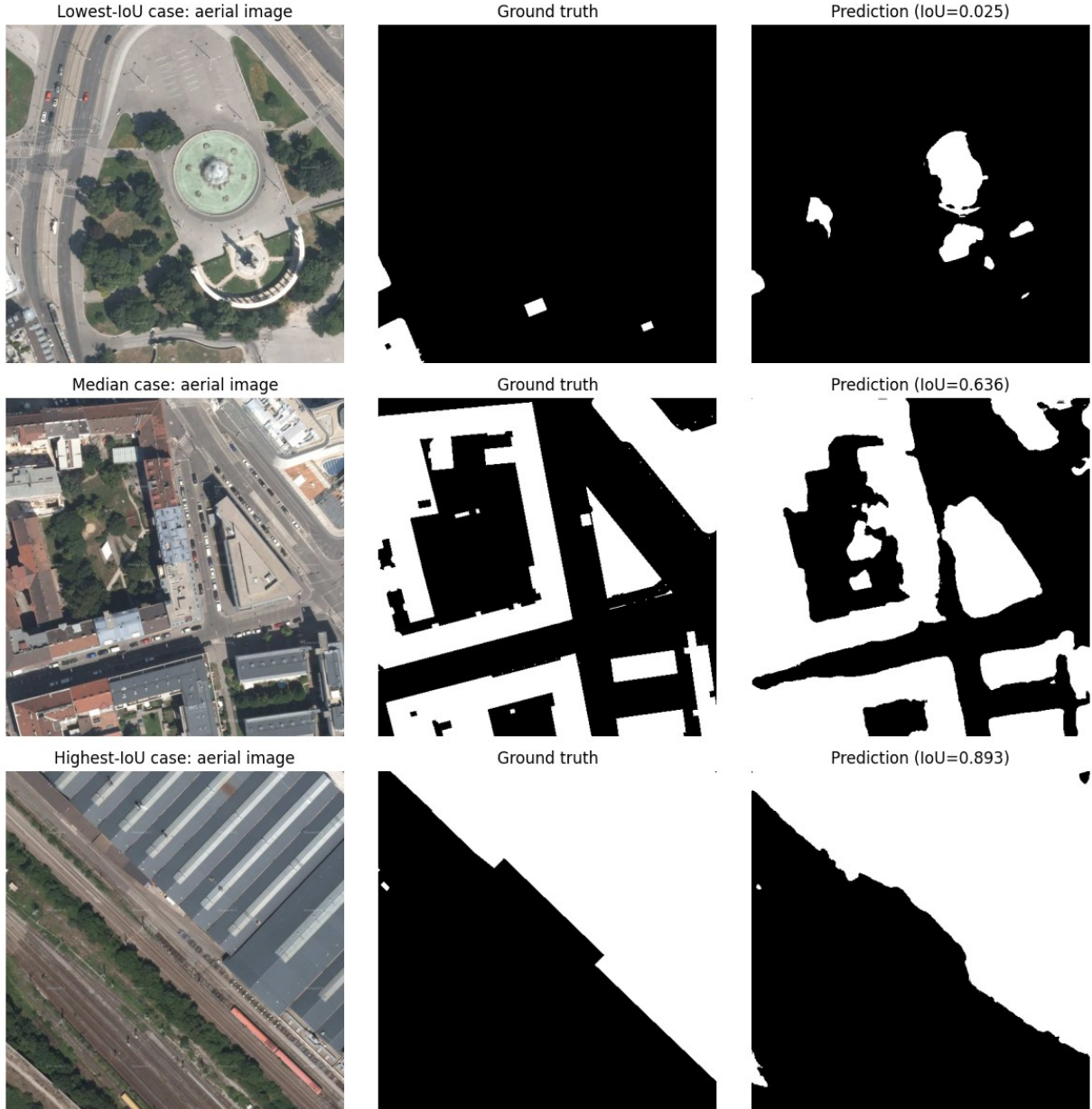


Figure 4. Original qualitative segmentation diagnostics from the deterministic notebook: lowest-, median-, and highest-IoU non-trivial Vienna patches. The selection criterion for visualization was at least 1% ground-truth building coverage.

5. Cross-Case Analysis

Table 6 maps the common evaluation logic to the two empirical cases. This table is central to the study because it connects the numerical results to the scope of the claims they can support.

Dimension	Case I: forecasting	Case II: segmentation
Deployment-relevant structure	Time: evaluation on later observations	Geography: evaluation on a held-out labeled city
Split unit	Monthly time window; chronological ordering	Entire city assigned before patch creation
Leakage guard	Evaluation labels never recycled into later fits; target components excluded as predictors	Vienna excluded from training/validation; city split precedes patch creation
Development evidence	Baseline comparison, walk-forward errors, Negative Binomial diagnostics	Validation on Tyrol-W; multi-metric held-out evaluation on Vienna
Key result	HGB-weather MAE 36.592; 52.6% below historical baseline	IoU 0.5876; Recall 0.8859; Boundary F1 0.2967
Failure/assumption exposed	Weather-enhanced result assumes ex-ante weather availability	False-positive over-segmentation and weak boundary localization
Strongest defensible claim	Performance under the stated 2011-2012 conservative temporal protocol	Performance on sampled patches from the author-defined held-out city of Vienna
Claim not supported	Magnitude of random-split optimism, universal future robustness, or causal weather effects	Official benchmark-test performance, universal cross-city robustness, or calibrated uncertainty

Table 6. Cross-case claim-evidence matrix with deliberately bounded inference.

RQ1 is addressed by the claim boundaries shown in Table 6. In forecasting, chronological separation supports a statement about performance under the stated late-month temporal protocol; in segmentation, city-level separation supports a statement about sampled held-out Vienna imagery. The common principle is therefore not that one split is universally correct, but that the unit of separation must match the structure of the deployment question before the resulting metric can be interpreted.

RQ2 is addressed descriptively with stronger preserved diagnostics in v2.3. In forecasting, the historical baseline comparison and the month-level MAE trajectory show predictive improvement under the stated protocol while exposing time-varying error and dependence on ex-ante weather. In segmentation, the all-patch and non-empty-patch summaries are similar, while Precision, Boundary F1, and the retained failure case expose false-positive and geometric weaknesses that are not evident from Dice alone. These diagnostics refine interpretation without creating unexecuted counterfactual claims.

Across the two cases, predictive performance and evidence about reliability are not equivalent. A low forecasting error or high segmentation recall does not show that a model will remain trustworthy under other future periods, cities, acquisition conditions, or data-generation regimes. The defensible statement depends on the holdout used, the assumptions required at prediction time, and the failure modes actually examined. This interpretation is consistent with prior work on performance and calibration degradation under dataset shift [2-4], while remaining narrower than a formal distribution-shift analysis.

6. Discussion

6.1 Evaluation design is part of the model claim

The central methodological observation is that the train-validation-test partition is part of the meaning of a reported performance number. A metric from a chronological holdout addresses a different prediction question from a metric obtained by randomly mixing observations across time, and a city-held-out segmentation result addresses a

different question from patch-level splitting within a shared geographic domain. Structured-validation literature supports this distinction [5,6,17-19]. The present experiments illustrate the distinction but do not quantify the performance difference versus random splits because that counterfactual was not preserved.

This point matters because applied ML studies can compress a complex deployment question into a single performance table. Here, the holdout structure is stated before interpreting the metrics. The resulting claims are intentionally narrow: the forecasting configurations are compared under one conservative 2011-2012 temporal protocol, and the segmentation configuration is evaluated on sampled patches from one author-defined held-out city. Neither claim is extended to untested future regimes, the Inria official blind test set, or other cities.

6.2 Aggregate metrics and hidden failure structure

The segmentation results illustrate why average overlap alone is insufficient. Dice of 0.7156 can look favorable in isolation, but the Precision-Recall imbalance and Boundary F1 show a substantial false-positive burden and weak edge localization. The non-empty-patch sensitivity analysis produces similar values, ruling out the simple explanation that good average overlap is mainly an artifact of empty-background patches. The qualitative triplets then show how those metric trade-offs appear in actual predictions. For mapping or engineering workflows that depend on geometry, these characteristics may matter as much as average overlap. Because no task-specific cost function or acceptance threshold was defined, the paper does not translate these metrics into an operational go/no-go judgment.

The forecasting case presents a different form of hidden dependency. The lowest observed errors are produced by a configuration that uses weather variables. Its retrospective performance is therefore conditional on those covariates being available with sufficient accuracy at prediction time. A deployment using actual weather forecasts would add a forecast-of-a-forecast uncertainty layer that is not represented here. The calendar-only model provides a reference with fewer ex-ante information requirements. The preserved outputs provide month-level MAE trajectories, but not confidence intervals or repeated-resampling uncertainty for those temporal windows.

6.3 Relationship to distribution-shift literature

The findings are interpreted in relation to, not as a new test of, the broader distribution-shift literature. WILDS documents natural shifts across locations, institutions, and time [2]; Taori et al. show that robustness to synthetic perturbations may not transfer to natural changes [4]; and Ovadia et al. show that uncertainty and calibration can deteriorate under shift [3]. The present study does not compute a formal divergence between development and holdout distributions. Accordingly, its empirical claims are framed in terms of structured temporal and geographic holdouts rather than measured shift magnitude.

The cross-case value is therefore a reporting and inference discipline: the holdout unit, leakage guard, assumptions, diagnostics, and maximum permitted claim are documented side by side for two modalities. This is useful as a transparent synthesis, but it should not be read as evidence that the framework itself is novel or that two cases are sufficient to establish a general law of machine-learning reliability.

6.4 Implications for engineering ML

Engineering applications can impose asymmetric costs for false negatives, false positives, uncertain boundaries, or delayed detection. A deployment-oriented study should therefore connect predictive outputs to explicit decision rules, acceptance thresholds, warning logic, abstention, fallback, or human review. The present paper does not evaluate such a decision layer; calibration and out-of-distribution methods [3,14,15] are identified only as candidate extensions.

A future engineering extension could evaluate railway perception under controlled and natural changes in lighting, weather, motion blur, camera configuration, infrastructure type, and geography. Such work would require pre-defined safety-relevant classes, repeated holdouts, calibrated uncertainty, and decision thresholds. No railway experiment or safety claim is made in the present manuscript.

7. Limitations and Reproducibility

- No random-split counterfactual is reported. The study therefore does not quantify performance optimism that might arise from random temporal or spatial splitting in these exact experiments.
- The forecasting model configurations are recoverable from the preserved notebook, but the pre-notebook history by which those hyperparameters were originally chosen is not documented. The manuscript therefore does not claim a pre-registered or unbiased global model-selection process.
- The forecasting protocol deliberately does not recycle evaluation labels into later fits. This creates strict separation but is not a standard expanding-window deployment simulation; a direct comparison with rolling or expanding origins remains necessary.
- The bikeshare data cover only 2011-2012. Observed weather fields are used as a proxy for covariates that would need to be known *ex ante*; operational forecasting should use genuine weather forecasts and propagate their uncertainty.
- Vienna is an author-defined held-out labeled city from the Inria data, not the benchmark's official blind test set. Only one final held-out city is evaluated, so broader geographic claims require rotating held-out cities and additional acquisition conditions.
- Each 5000 x 5000 Inria tile contributes 16 non-overlapping 512 x 512 patches at starts 0, 1496, 2992, and 4488 pixels on each axis. This samples about 16.8% of tile area, so the experiment is not a full-tile evaluation and may miss unsampled local structure.
- The segmentation standard deviations are patch-level dispersion, not confidence intervals for city-level generalization. Spatial dependence means 576 patches should not be interpreted as 576 independent geographic replicates.
- The segmentation result is a deterministic single-seed execution, not a multi-seed uncertainty study. Deterministic seed control improves exact repeatability of the run but does not measure sensitivity to alternative initializations; repeated independent seeds remain necessary.
- The two cases are intentionally heterogeneous, so their metric values cannot be pooled into a common statistical effect. Cross-case conclusions are analytic and methodological rather than meta-analytic.
- Version 2.3 links this manuscript to a public reproducibility repository containing a deterministic notebook and Python export with public-data download logic, exact patch coordinates, model settings, seed controls, and preserved outputs. The repository URL is reported in the Code availability statement. Exact package versions from the original 2026 Colab runtime were not archived, so environment-level reproducibility remains incomplete.

8. Conclusion

This research paper examines how the structure of a holdout constrains the generalization statement that can be supported in two applied machine-learning settings. The preserved evidence does not establish a universal robustness method, quantify random-split optimism, or measure formal distributional divergence. It shows instead that temporal and geographic holdouts produce claims whose scope must be stated together with the protocol that generated them.

Under the stated conservative temporal protocol, HGB-weather produced the lowest observed pooled errors among the reported forecasting configurations and a 52.6% lower MAE than the historical baseline; the preserved monthly trajectory shows lower HGB-weather MAE than the baseline in every plotted evaluation window, while errors vary over time. In the segmentation case, the deterministic U-Net/ResNet-34 execution produced mean IoU 0.5876 and Recall 0.8859 on 576 sampled held-out Vienna patches, while Precision 0.6345 and Boundary F1 0.2967 exposed false-positive and boundary-localization weaknesses. Similar metrics on the 543 non-empty patches and retained qualitative failure cases reinforce that interpretation. These results remain informative only within the stated evaluation boundaries.

The practical implication is that predictive metrics should be accompanied by an explicit statement of the holdout unit, information excluded from development, assumptions required at prediction time, random-state controls, the uncertainty represented by reported statistics, and claims that remain unsupported. The next empirical priority is a

pre-specified comparison of random versus deployment-aligned splits, repeated temporal protocols, rotating geographic holdouts, multiple independent training seeds, calibrated uncertainty, and full environment capture.

Declarations

Generative AI disclosure. During preparation of this manuscript, the author used OpenAI ChatGPT to assist with literature organization, language refinement, document restructuring, and drafting based on experimental outputs and source material supplied or verified by the author. The author reviewed, edited, and verified the resulting text, numerical results, calculations, citations, interpretations, and conclusions and takes full responsibility for the content.

Author affiliation note. The author is professionally affiliated with Usinas Siderurgicas de Minas Gerais S.A. - Usiminas. This research was conducted independently; the views expressed are solely those of the author and do not represent or imply endorsement by the employer.

Data availability. The Capital Bikeshare / Bike Sharing dataset and the Inria Aerial Image Labeling dataset are publicly available from the original repositories and sources cited below. The segmentation results in this paper use an author-defined split of labeled Inria cities rather than the official blind benchmark test set.

Code availability. The public reproducibility package for the computational workflow used in this manuscript is available at <https://github.com/Herislei/deployment-aligned-evaluation-v2.2>. The repository contains the deterministic notebook, plain-Python export, README documentation, public-data download logic, forecasting configurations, author-defined city split, exact patch-coordinate generation, PyTorch/CUDA seed controls, evaluation definitions, and preserved diagnostic outputs used to support the reported results. The package does not archive the exact package versions of the original 2026 Colab runtime, so complete environment-level reproduction is not claimed. No random-split, rotating-city, or multi-seed results are claimed in this revision.

Competing interests. The author declares no competing interests related to this manuscript.

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